WASTEWISE SCAN · DESIGN DECISION RECORD

WasteWise Scan shared scan record format

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DOCUMENT CONTROL

Document ID	WSC-DDR-002	Version	0.4
Status	Draft	Date	2026-10-02
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/wastewise-scan		
Source	docs/decisions/0002-scan-record-format.md @ 8c61176		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-02	Amish Chadha	Pair ID screen (QR code and digit fallback) added to the record fields and to the firmware sketch notes, carrying the decision of 2026-10-02
0.3	2026-10-02	Amish Chadha	Field questions 2 and 3 recorded as decided by Amish on 2026-10-02
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); option 1 decided for WasteWise Scan (WSC-DDR-003 D12); WasteWise-ml adoption listed as a cross-repo action
0.1	2026-09-25	Amish Chadha	First proposal of a scan record shared with WasteWise-ml, for joint review

0002: Shared scan record format

- **Date:** 2026-09-25
- **Status:** accepted for WasteWise Scan (decided by Amish, 2026-09-25: go with recommendation). Adoption by WasteWise-ml is a cross-repo action; field questions 2 and 3 below were decided by Amish on 2026-10-02 (WSC-DEC-001).

Context

Requirement R13 asks that low-confidence scans hand off to WasteWise-ml and that scan records use a shared, documented format. WasteWise-ml labels items with the taxonomy in its `ml/data/taxonomy.yaml` (version 0.2, 2026-09-25): a material class, a grade (for plastics, the resin code, for example "PET (1)" or "other or unknown (7)"), and two non-class outputs, "unsure" and "hazard". WasteWise Scan produces a resin, a confidence, a local price grade and an eight-band spectrum. ReflowEconomy will later want the resin of what it buys.

This note is written in the WasteWise Scan repo only. The WasteWise-ml repo was read and not changed. Amish accepted option 1 on 2026-09-25; the WasteWise-ml side has still to adopt the same fields in its own repo.

Options considered

1. **One shared record for both tools (recommended).** Both write the same fields; each leaves the fields it cannot fill empty. Combining a photo and a scan of the same item is a join on `pair_id`.
2. **Separate records with a mapping table.** Each tool keeps its own format and a converter joins them. Less coordination now, more drift later.
3. **Adopt an external schema.** No open schema for field resin scans was found in this session; this can be revisited.

Proposed record

Recommendation: option 1, with the fields in Table 1. Exports are CSV (one row per record, spectrum as eight columns) and JSON Lines (one object per line). Labels reuse the WasteWise-ml taxonomy strings exactly, so no mapping is needed.

Table 1. Shared scan record, version 0.1.

Field	Type	Filled by	Meaning
`record_version`	text	both	"0.1"
`source`	text	both	"wws" (WasteWise Scan) or "wml" (WasteWise-ml)
`device_id`	text	both	Short random ID set at first start; no personal data
`record_id`	text	both	`device_id` plus a sequence number, unique per device
`time_utc`	text	both	ISO 8601, UTC, to the second
`site_id`	text	both, optional	The price table in use (cooperative or buyer site); set by the partner
`pair_id`	text	both, optional	Same value on a photo and a scan of the same item; nine digits made by the scanner, shown on its screen as a QR code and in digits, and scanned or typed on the phone; empty if unpaired
`pair_via`	text	both, optional	How the phone received the pair ID: "qr" or "typed"; empty if unpaired
`material_class`	text	both	Taxonomy class; WasteWise Scan writes "plastic" or leaves it empty for "unknown"
`grade`	text	both	Taxonomy grade, for example "PET (1)", "PP (5)", "other or unknown (7)"
`output`	text	both	"answer", "unsure" or "hazard"; WasteWise Scan never writes "hazard"
`confidence`	number	both	0 to 1
`price_grade`	text	both, optional	Local grade from the site price table, for example "A"
`model_version`	text	both	Classifier or model version
`wws_bands_nm`	list of 8 numbers	WasteWise Scan	850, 940, 1050, 1200, 1300, 1450, 1550, 1650
`wws_reflectance`	list of 8 numbers	WasteWise Scan	Dark-subtracted signal divided by the last white reference
`wws_dark_v`	number	WasteWise Scan	Mean dark level, a sunlight indicator (see WSC-CAL-001, D)
`wws_cal_id`, `wws_temp_c`	number	WasteWise Scan	White reference used, and board temperature
`wml_image_ref`	text	WasteWise-ml	File name of the consented photo, if kept

On the scanner, each record is stored in a fixed 64-byte binary form (time, sequence, result code, confidence, price grade, flags, pair ID, eight 32-bit reflectance values, dark level, calibration ID, temperature, classifier version and a CRC-16). WSC-CAL-001 section G sizes the log from this. The CSV and JSON Lines exports are generated from it.

Decision

Decided by Amish, 2026-09-25: go with recommendation. WasteWise Scan uses option 1, one shared record with the fields in Table 1 and a `pair_id` join (recorded in WSC-DDR-003 D12). Questions left open on 2026-09-25, with no recommendation made then:

1. Whether WasteWise-ml accepts the shared field names and the `pair_id` join (cross-repo action; the WasteWise-ml side decides in its own repo).
2. How `pair_id` is created in the field: typed digits, a sticker, or the phone scanning a code on the scanner screen. Decided by Amish, 2026-10-02: the scanner creates the pair ID and shows it as a QR code on its screen for the phone to scan, with the same short code in digits as a typed fallback.
3. Whether spectra from consented scans join the WasteWise-ml field dataset, and under which license and ownership terms (WasteWise-ml R14). Decided by Amish, 2026-10-02: consented spectra join the WasteWise-ml field dataset on the same terms as field photos: opt-in per scan, co-owned by the pickers' organization, and published under CC BY 4.0 only with that organization's agreement.

Pair ID on the scanner screen

Decided by Amish, 2026-10-02 (question 2). These are notes for the firmware sketch; no firmware is written at TRL 3.

- After a scan the screen offers the pair ID. A short press on the button shows a screen with a QR code on the upper part and the same code in digits under it, in three groups of three (for example 482 913 057), large enough to read at arm's length.
- The scanner makes the pair ID when the result is stored: nine digits from its random generator, never from the time or the device, so the code carries no personal data. The same nine digits go into the stored record's pair ID field, so the scan and the code on the screen are the same string.
- The QR code holds the text `WWS1-<nine digits>-<device_id>`. At about 20 characters (a device ID of up to eight characters) it fits the smallest versions of the code with a wide margin (version 2 holds 47 alphanumeric characters at the lowest error correction). At version 2 the code is 25 modules wide; with four modules of quiet zone each side and 5 pixels a module it is 165 pixels wide, which fits the 170 pixel width of the display board, and it is drawn black on white at full backlight so a phone reads it in sun.
- The phone side (WasteWise-ml) reads the QR code to fill the pair ID and the device ID, or the user types the nine digits and chooses the scanner. The record then stores `pair_via` as "qr" or "typed" (on the scanner it is one bit of the flags byte, so the 64-byte record does not grow).
- Nine digits give a billion codes. A join also needs the same `device_id` and a photo within 24 hours of the scan, so a chance clash between two items is negligible even at ten thousand pairs on one site.
- The code stays on screen until the next scan or a second button press, and is not re-made for the same scan; a typed code that matches no scan is held for 24 hours and then marked unpaired.
- A scan can still be stored with no pair; the screen is optional and never delays the next scan.

Consequences

- WasteWise-ml is asked to add the same fields on its side, in its own repo and session (cross-repo action in `docs/REVIEW.md`).
- R13 stays "at risk" in WSC-CAL-001 until WasteWise-ml adopts the format.